Code No: 115AP **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, March - 2024
COMPILER DESIGN
(Computer Science and Engineering)

Time: 3 hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART - A

(25 Marks)

1.a) Define lexeme, token, pattern. [2]
b) Construct the transition diagram for relational operators in 'C'. [3]
c) What is meant by goto function in LR parser? Give an example. [2]
d) For the grammar S->0S1 | 01. Indicate the handle in each of the following right-sentential forms: i) 000111 ii) 00S11 [3]
e) Write SDTs for the Boolean expression grammar [2]
f) List the properties of a Memory manager for efficient usage of memory. [3]
g) What is a DAG? [2]
h) Define flow graph and provide an example. [3]
i) What is machine dependent optimization? [2]
j) How would you calculate the cost of an instruction with an example? [3]

PART - B

(50 Marks)

2.a) Explain the various phases of a compiler. Show the translations for an assignment statement position=initial+rate *60, clearly indicate the output of each phase.
b) Explain the concept of bootstrapping in the context of compilation, discussing its significance and potential challenges. [5+5]

OR

3.a) Construct recursive descent parsing for the following grammar.

S→cAd
A→ab | a

b) Construct predictive parsing table for the following grammar:
S → (L) | a L → L, S |S
Check whether the string (*a*, ((*a* , *a*), (*a* , *a*))) belongs to that language or not. [5+5]

4.a) Explain shift-reduce parsing with stack implementation.
b) Build LR(1) parser and check the validity of the input string *"id+id*id"* by the LR(1) parser for the given grammar
E →E+T/T, T→T*F/F, F→ (E)/id [5+5]

OR

5.a) Discuss the challenges associated with error recovery in parsing.

b) Construct an LALR parsing table for the following grammar: [5+5]

$$S \rightarrow CC,\ C \rightarrow cC,\ C \rightarrow d$$

6.a) For the grammar below: E → E + T | T, T → num .num | num Give an SDD to determine the type of each term T and expression E.

b) Write the translation scheme for type checking of function declaration. [5+5]

OR

7.a) Explain tree structured symbol tables.

b) Explain in detail, the strategy for reducing fragmentation in heap memory. [5+5]

8.a) Discuss about principal sources of optimization with example.

b) Write a short note on following optimizations concepts:
i) Frequency reduction ii) Folding. [5+5]

OR

9.a) Generate the flow-graphs for the following expressions:

S-> id: = E |S; S| if E then S else S | do S while E
E-> id + id |id

b) Explain about common sub expression elimination with an example. [5+5]

10.a) Discuss the issues in code generation.

b) Discuss about register allocation and assignment in target code generation. [5+5]

OR

11.a) What is peephole optimization? How can it be performed? Give its role in code generation.

b) Write an algorithm for generating code from DAG with an example. [5+5]

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